Teaching Coding Agents to Master *Spreadsheets*

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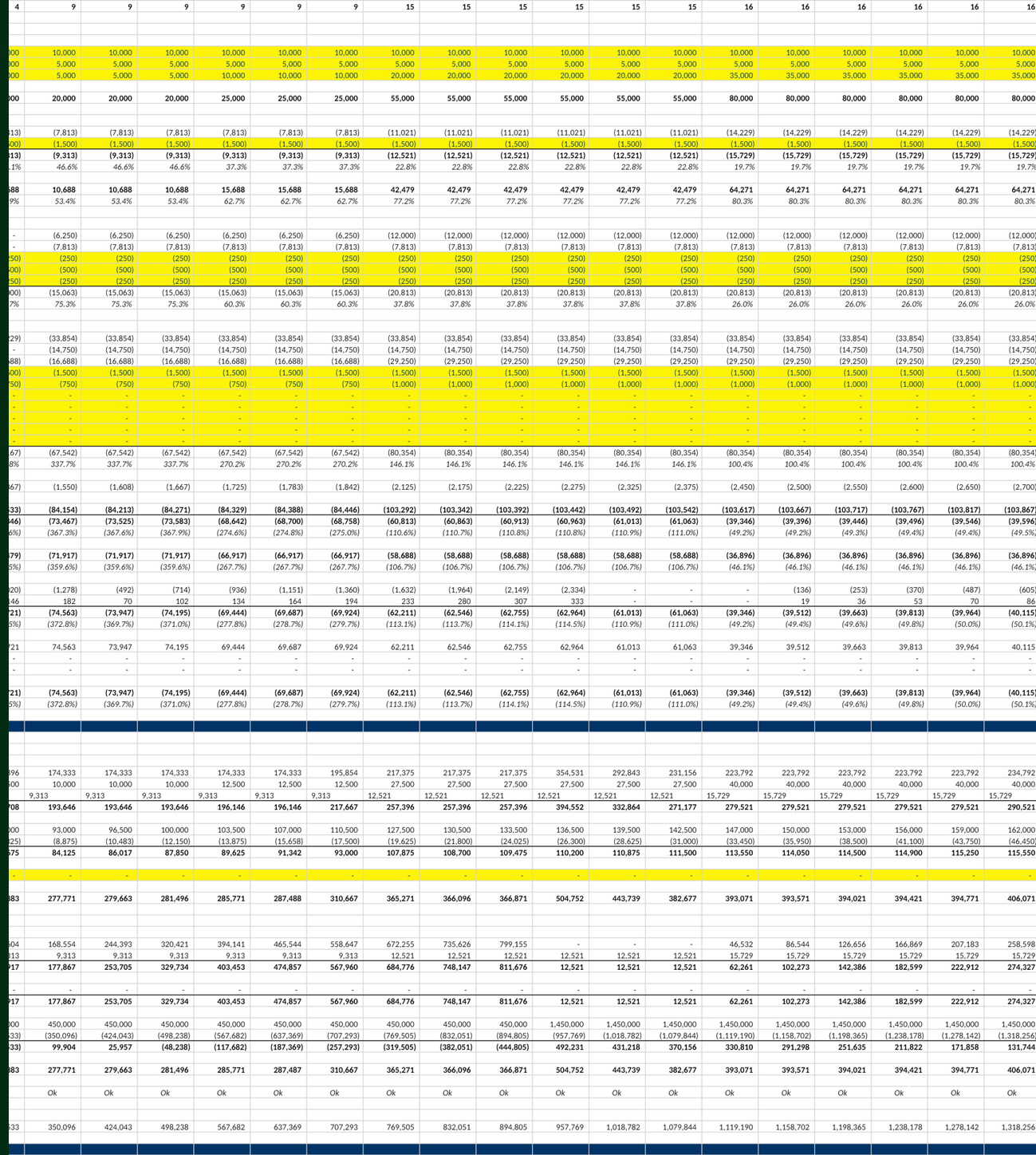
50% → 92%

- 4 months, multiple architectures, and many dead ends
- What mattered most: replacing 15 discrete tools with one REPL

The problem

A human sees: revenue table, assumptions, P&L summary

An LLM sees: 10,000 cell values, `=SUMPRODUCT((B$3:B$50="NEW")*(G$3:G$50))`, formatting metadata



Three specialized agents:

1. Block discovery — identifies workbook structure
2. Edit agent — 5-step process: disambiguate, define end state, plan, execute, verify
3. Question agent — answers questions

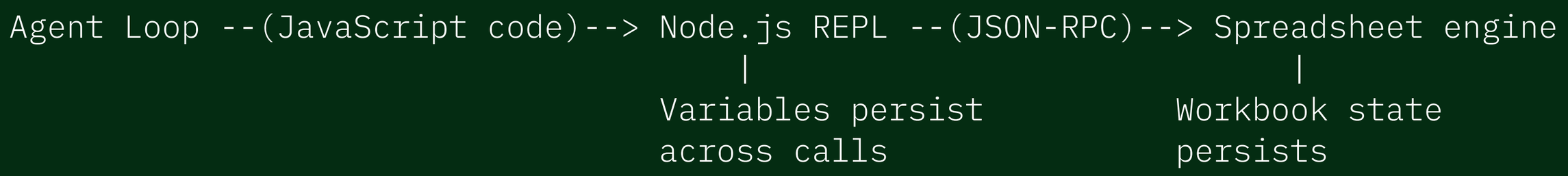
Key finding: Rigid architectures don't win

More dead ends

Representation	Why it failed
TSV views	Lost formatting and structural context
SQL views	Flat tables didn't capture visual structure
HTML tables	Too verbose, consumed too many tokens
DOT graphs	Useful for analysis, not for interaction
XML/XSLT	LLM struggled with the syntax

None of them worked as a general-purpose representation — but two informed what came next.

We replaced all 15 tools with a persistent Node.js REPL and a spreadsheet API.



Before vs. After

Before (10-15 tool calls):

```
Tool call 1: list_sheets()              -> ["Summary", "Data", "Inputs"]
Tool call 2: read_range("Summary!A1:D10") -> cell values
Tool call 3: find_cells("Revenue")    -> [matches]
Tool call 4: read_cell("Summary!C10") -> "$1,234,567"
...
```

After (1 tool call):

```
const sheets = await xlsx.listSheets(wb);
const summary = await xlsx.readRangeTsv(wb, `${sheets[0].name}!A1:D10`);
const revenue = await xlsx.findCells(wb, "Revenue", { context: 1 });
console.log(sheets, summary, revenue);
```

Code mode vs. REPL

Code mode — the agent writes scripts instead of making tool calls. Operations compose naturally. 50-line scripts are common.

REPL = code mode + persistent state — variables survive across calls. The agent writes shorter scripts, reasons between them, builds understanding incrementally.

```
// Code mode: one long script, print everything at the end
// REPL: explore, reason, continue
const revenue = await xlsx.findCells(wb, "Revenue", { context: 1 });
console.log(revenue);
// → agent reasons about the output, then writes the next script
```

Result: accuracy gain on harder tasks — the agent can course-correct mid-exploration.

The results

Date	Pass rate	Tasks
Nov 30	**74%**	104
Dec 11	**88%**	167
Dec 14	**92.1%**	165

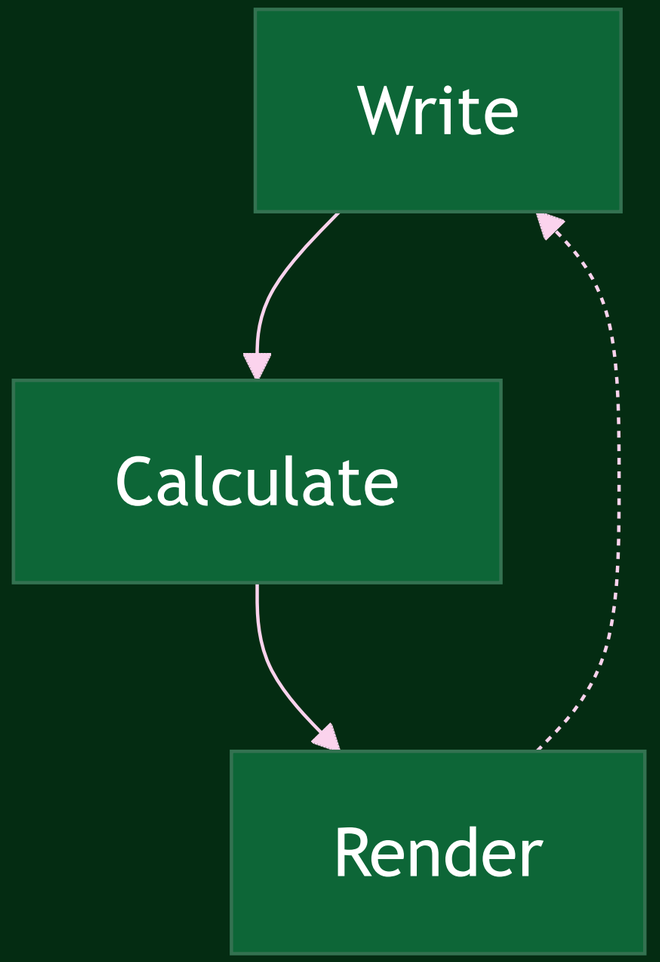
Zero timeouts. 50-second average runtime.

18 points in two weeks from compounding gains:
better search, new API methods, improved docs, backend bug fixes

The verification loop

The formula engine and renderer close a loop the agent uses to check its own work.

This held across three successive frontier model releases. Each new model used the same loop more effectively.



The **REPL** is an interface — the best one today, because coding is where models are strongest.

The **engines** — formula calculation, rendering, linting — are the more durable part. They're what close the verification loop, and they compound with each new model.

If for instance agents become as capable at computer use as they are at coding, the interface might change. The engines won't.

Domain knowledge outlived every tool

- We changed tools four times in four months
- The financial domain knowledge improved results on **every one of them**

Structured as a composable prompt component:

- How to interpret margins, profitability, revenue cascades
- Model type recognition (DCF, LBO, three-statement)
- Communication conventions ($1.2M not $1,234,567.89)
- "Never calculate in your head what the spreadsheet can calculate for you"

It was the most reused component in the system.

Evaluation was half the work

We started with only LLM-as-judge. We couldn't tell if a score change was the agent improving or the evaluator being flaky.

We replaced it with deterministic comparison wherever possible — programmatic checks for values, layout, and formatting. LLM grading only for genuinely subjective text answers.

568 commits. Nearly as complex as the agent itself.

What generalizes

1. **Many small sequential tool calls = a bad scripting language.** Give the agent a real one.
2. **Build verification engines.** Formula calc, rendering, and linting close a feedback loop that compounds with model capability rather than being replaced by it.
3. **Interfaces are ephemeral, engines are durable.** The REPL works because coding is today's strongest model skill. That may or may not last.
4. **Domain knowledge outlasts tools.** Four backends in four months. The domain knowledge improved results on all of them.
5. **Match evaluation to output type.** Deterministic comparison for objective outputs, LLM grading for subjective ones.
6. **Check the plumbing first.** Agent "confusion" is usually infrastructure.

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